MATH 1342 – Elementary Statistics – Week 12: Inference for Two Proportions and for a Mean

Nov 10 – 12 | 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) • 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CLO 5) **2.** Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **3.** Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT) Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. Explain why inference for a mean uses the t distribution with df = n − 1, and check the conditions for t procedures. (CLO 5) **5.** Construct and interpret a t confidence interval for a population mean, and carry out a one-sample t test. (CLO 6, CLO 8)	e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT) Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 5.1, 6.1–6.3 (all objectives)	Assignment Tool in Blackboard

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.